

Technical Appendix

Cyclistic Bike-Share Case Study — Data Cleaning & Processing Documentation

This appendix documents the full technical methodology behind the Cyclistic case study analysis, including tools, formulas, validation checks, and issues encountered and resolved during data processing. It is intended for technical reviewers and accompanies the main stakeholder report.

1. Tools & Environment

- Microsoft Excel 2016 — used for manual cleaning of an initial subset of monthly files, to build and validate the cleaning methodology.
- Python 3 (pandas) in Jupyter Notebook — used to automate the identical methodology across all twelve monthly files for consistency and scale.
- SQLite (DB Browser) — used once as a workaround for an Excel performance issue on a large-scale row deletion (see Section 4).

2. Data Preparation

Each monthly raw CSV file (ride_id, rideable_type, started_at, ended_at, start/end station name & id, start/end lat/long, member_casual) was downloaded from Divvy's public trip-data repository and stored in a dedicated raw-data folder, preserving an untouched original copy per standard data-hygiene practice.

3. Calculated Fields

Ride length

```
Excel: =D2-C2 (ended_at - started_at)
Python: df["ride_length"] = df["ended_at"] - df["started_at"]
```

Day of week (Divvy convention: 1=Sunday ... 7=Saturday)

```
Excel: =WEEKDAY(C2,1)
Python:
def to_divvy_weekday(dt_series):
    pandas_dow = dt_series.dt.dayofweek # Mon=0 ... Sun=6
    return ((pandas_dow + 1) % 7) + 1 # Sun=1 ... Sat=7
```

Both implementations were cross-validated: Excel formulas were checked against known calendar dates (e.g., confirming July 5, 2025 = Saturday = 7), and the Python function was validated by comparing automated cleaning results for July and August against the independently-produced manual Excel results (see Section 6).

4. Data Quality Checks & Exclusion Rules

Three outlier categories were checked for and, where applicable, excluded, based on Divvy's own documented data-quality guidance:

Check	Rule	Rationale
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Negative duration	ride_length < 0	Data error (ended before started); none found across any month.
Excessively long rides	ride_length > 24h	Predominantly reflects lost/stolen bikes rather than genuine trips; system-forced trip terminations were observed clustering near a fixed ~24h59m mark, supporting this interpretation.
Excessively short rides	ride_length < 60s	"False starts" per the data provider's own documentation.

Excel helper column:

```
=IF(OR(N2>1, N2<TIME(0,1,0)), "DELETE", "KEEP")
```

Python equivalent:

```
to_delete = (df["ride_length"] > pd.Timedelta(days=1)) | (df["ride_length"] <
pd.Timedelta(minutes=1))
cleaned_df = df.loc[~to_delete]
```

Missing end-station coordinates (end_lat/end_lng) were also checked and logged (typically under 0.15% of rows per month) but retained, since location data was not required for the ride-length / day-of-week analysis.

5. Issues Encountered & Resolutions

5.1 Formula vs. static value confusion during bulk fill

An early attempt to fill formulas across ~760,000 rows via copy/paste resulted in a single static value being repeated across all rows, rather than a live formula. Diagnosed by inspecting the Formula Bar for a distant row; resolved by re-copying and using Paste Special > Formulas, then converting to static values only once row deletion was imminent.

5.2 Scientific-notation corruption of ride_id

Approximately 533 ride_id values in one monthly Excel file were silently auto-converted by Excel from alphanumeric text to scientific-notation numbers (e.g., 6.78315E+15) the first time the raw CSV was opened, causing irreversible precision loss. Root cause: Excel's automatic type inference on alphanumeric ID columns. Resolution: since ride_id was not used as an analytical key, affected rows were retained with the ride_id column flagged as unreliable for that subset. Preventive practice for future files: force a Text column type on ID fields during CSV import (Data > From Text/CSV) rather than opening the file directly. Python's CSV reader, by contrast, does not exhibit this issue, since it reads such fields as strings by default.

5.3 Date component lost during a tool hand-off (Excel to SQLite to CSV)

After using SQLite to complete a large-scale row deletion that Excel could not perform without freezing, the exported CSV's started_at/ended_at columns retained only the time component, having silently lost the date. This was diagnosed by comparing row counts and column types against the pre-deletion Excel file, then resolved by re-merging the correct full datetime values back from that file via a ride_id-based lookup. Preventive practice: verify timestamp integrity immediately after any cross-tool export/import.

5.4 Display truncation vs. genuine data errors

On multiple occasions, a duration value appeared visually implausible (e.g., a truncated "25:47.1") purely due to Excel column-width/format display truncation, not a genuine data problem. This was confirmed each time using a helper formula such as =TEXT(cell,"yyyy-mm-dd hh:mm:ss"), which forces Excel to reveal the true

underlying stored value — a general-purpose technique for distinguishing display artifacts from real data-integrity issues.

5.5 Column-size limitations of spreadsheet tools at scale

Excel's 1,048,576-row-per-sheet limit made it unsuitable for opening the final combined 12-month dataset (5,764,564 rows), producing a "file not loaded completely" warning and application freezes. This was resolved by continuing all further exploratory analysis directly in Python (pandas), which loaded and processed the full dataset in seconds — illustrating a practical decision point between spreadsheet and programmatic tools based on data scale.

6. Cross-Validation Between Manual and Automated Cleaning

As a methodology sanity check, results from manually-cleaned Excel files (July and August) were compared against the automated Python pipeline applied to the same raw source files:

Month	Manual (Excel) result	Automated (Python) result	Match
July 2025	738,825 rows	738,824 rows	Match (1-row rounding difference, immaterial)
August 2025	766,537 rows	766,537 rows	Exact match

This close agreement between two independently-implemented methodologies provides strong confidence that the cleaning logic is correct and consistently applied, rather than a coincidental or tool-specific result.

7. Final Dataset

- Combined 12-month dataset: 5,764,564 valid ride records (July 2025 - June 2026).
- Standard schema retained: ride_id, rideable_type, started_at, ended_at, start/end station name & id, start/end lat/lng, member_casual, ride_length, day_of_week.
- Export format optimized (timedelta and datetime fields converted to compact string representations) to reduce file size prior to final export.